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Active Cryptographic Tissue-Impedance Watermarking for Hardware-Root-of-Trust Biometric Authentication

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Abstract*— *Biometric authentication relies on Presentation Attack Detection (PAD) to verify a real body part is placed on a sensor. A distinct threat bypasses the sensor entirely: synthetic or stolen biometric data is injected directly into the path between sensor and processor, a category formalized as Biometric Data Injection Attacks, against which PAD offers no protection.

As a countermeasure a low-energy signal is passed through the user's tissue via electrodes integrated with the sensor, using the body's natural bioelectrical response to cryptographically bind the captured biometric sample to the physical act of capture. Since this binding can only be produced by live tissue at the sensor at that instant, an injected sample cannot satisfy it.

I. Introduction

Biometric authentication is the default trust mechanism for smartphones, payments, and restricted-access infrastructure. For over a decade the primary defense against spoofing has been PAD (ISO/IEC 30107), verifying a real body part is presented to the sensor. PAD only inspects the sensor surface and assumes data reaching the processor afterward is what the sensor captured.

Biometric data injection attacks break this assumption by inserting synthetic or stolen data directly into the internal communication path. This work proposes binding the biometric sample to a signal only live tissue in contact with the sensor can produce, so an injected sample cannot satisfy authentication without a real, present user.

II. Methods

The proposed system, illustrated in **Fig. 1**, integrates a bioelectrical liveness mechanism directly into the biometric sensor assembly. At a conceptual level, the design consists of four functional stages.

A low-energy signal is passed through the user's tissue via contact electrodes positioned around the biometric sensor. As this signal propagates through living tissue, it is naturally and measurably altered by the body's bioelectrical characteristics in a way that is difficult to reproduce without a real, live body in contact with the electrodes. The altered response is captured and checked for consistency with expected biological characteristics, including its variation over time, before being accepted as a valid liveness indicator.

Rather than treating this liveness check and the biometric match as independent conditions, the verified response is used to cryptographically bind the biometric sample to the specific physical capture event, producing a single artifact that reflects both. This is a deliberate design choice: independent checks can, in principle, be satisfied separately by a live but unauthorized user and an injected sample presented concurrently, whereas a bound artifact requires both conditions to originate from the same physical event. Final access is granted only when this bound artifact is successfully verified at the controller, so that a mismatch in either the biometric content or the physical origin of the signal results in rejection.

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III. Results and Discussions

As this is a proposed system, evaluation is planned rather than completed. Testing will focus on three things: false rejection rate across varying skin and environmental conditions, resistance to simple electrical bypass attempts (e.g., static components mimicking tissue), and correct rejection of injected biometric samples that lack a valid live signal. The design is expected to raise the cost and effort required for an attack, not eliminate risk entirely. No physical security mechanism offers absolute guarantees.

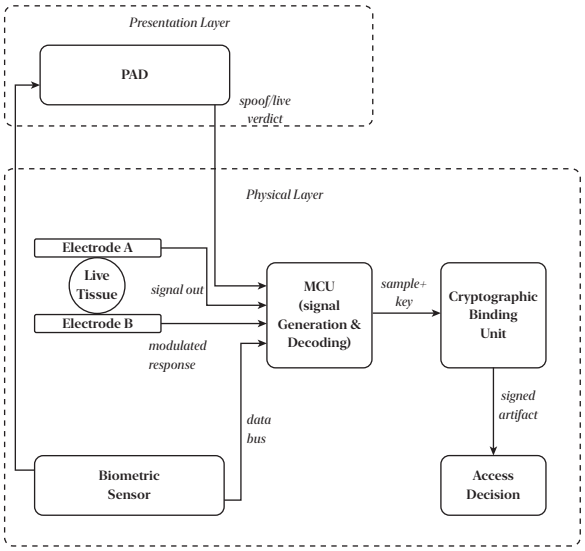


Fig. 1 Wire-level signal flow from live tissue to access decision

IV. Conclusion

Presentation Attack Detection addresses spoofing at the sensor but was never designed to address data injected after the sensor is bypassed. The proposed mechanism closes part of this gap by binding the biometric sample to a signal that only a live, present body can produce, complementing rather than replacing existing defenses. The scope and limitations discussed above are intended to state honestly where the design's guarantees currently end, and the described concept is intended for narrow, high-assurance deployments such as restricted facility access rather than general consumer use.

References

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